Exercise III

Registration No. : 2025/CS/079
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(01)

(a)

```c
#include <stdio.h>

int main() {
    int n = 14;

    if (n % 2 == 0) {
        printf("n is even\n");
    }

    return 0;
}
```

```
chamal@MacBook-Air: ~/Desktop $ ./a.out
n is even
chamal@MacBook-Air: ~/Desktop $
```

(b)

```c
#include <stdio.h>

int main() {
    int n = 13;

    if (n % 2 == 0) {
        printf("n is even\n");
    } else {
        printf("n is odd\n");
    }

    return 0;
}
```

```
chamal@MacBook-Air: ~/Desktop $ ./a.out
n is odd
chamal@MacBook-Air: ~/Desktop $
```

(c)

```c
#include <stdio.h>

int main() {
    char c = 'C';

    if (c > 64 && c < 90) {
        printf("c is in upper case\n");
    } else if (c > 97 && c < 122) {
        printf("c is in lower case\n");
    }

    return 0;
}
```

```
chamal@MacBook-Air: ~/Desktop $ ./a.out
c is in upper case
chamal@MacBook-Air: ~/Desktop $
```

(d)

```c
#include <stdio.h>

int main() {
    char c = 'z';

    if (c > 64 && c < 91) {
        printf("c is in upper case\n");
        printf("lower case c is %c\n", c + 32);
    } else if (c > 96 && c < 123) {
        printf("c is in lower case\n");
        printf("upper case c is %c\n", c - 32);
    }

    return 0;
}
```

```
chamal@MacBook-Air: ~/Desktop $ ./a.out
c is in lower case
upper case c is Z
chamal@MacBook-Air: ~/Desktop $
```

(e)

```c
#include <stdio.h>

int main() {
    char c = '5';

    if (c > 64 && c < 91) {
        printf("c is in upper case\n");
        printf("lower case c is %c\n", c + 32);
    } else if (c > 96 && c < 123) {
        printf("c is in lower case\n");
        printf("upper case c is %c\n", c - 32);
    } else if (c > 47 && c < 58) {
        printf("numbers\n");
    } else {
        printf("Others\n");
    }

    return 0;
}
```

```
chamal@MacBook-Air: ~/Desktop $ ./a.out
numbers
chamal@MacBook-Air: ~/Desktop $
```

(02)

(a)

```c
#include <stdio.h>

int main() {
    int num1 = 10;
    int num2 = 5;
    int answer;
    char oper = '*';

    switch (oper) {
        case '+' : answer = num1 + num2; break;
        case '-' : answer = num1 - num2; break;
        case '/' : answer = num1 / num2; break;
        case '*' : answer = num1 * num2; break;
        case '%' : answer = num1 % num2; break;
    }

    printf("%i %c %i = %i\n", num1, oper, num2, answer);

    return 0;
}
```

```
chamal@MacBook-Air: ~/Desktop $ ./a.out
10 * 5 = 50
chamal@MacBook-Air: ~/Desktop $
```

(b)

```c
#include <stdio.h>

int main() {
    int num = 999, len = 0, n;
    len = (num <= 9) ? 1 : (num <= 99) ? 2 : (num <= 999) ? 3 : 0 ;

    switch (len) {
        case 3:
            n = num;
            switch (n) {
                case 100 ... 199 :
                    printf("One Hundred ");
                    n = num % 100;
                    goto twodigit;
                    break;
                case 200 ... 299 :
                    printf("Two Hundred ");
                    n = num % 100;
                    goto twodigit;
                    break;
                case 300 ... 399 :
                    printf("Three Hundred ");
                    n = num % 100;
                    goto twodigit;
                    break;
                case 400 ... 499 :
                    printf("Four Hundred ");
                    n = num % 100;
                    goto twodigit;
                    break;
                case 500 ... 599 :
                    printf("Five Hundred ");
                    n = num % 100;
                    goto twodigit;
                    break;
                case 600 ... 699 :
                    printf("Six Hundred ");
                    n = num % 100;
                    goto twodigit;
```

```
chamal@MacBook-Air: ~/Desktop $ ./a.out
Nine Hundred Ninety Nine
chamal@MacBook-Air: ~/Desktop $
```

```c
                    break;
                case 700 ... 799 :
                    printf("Seven Hundred ");
                    n = num % 100;
                    goto twodigit;
                    break;
                case 800 ... 899 :
                    printf("Eight Hundred ");
                    n = num % 100;
                    goto twodigit;
                    break;
                case 900 ... 999 :
                    printf("Nine Hundred ");
                    n = num % 100;
                    goto twodigit;
                    break;
            }
        case 2:
            n = num;
twodigit:
            switch (n) {
                case 10 : printf("Ten\n"); break;
                case 11 : printf("Eleven\n"); break;
                case 12 : printf("Twelve\n"); break;
                case 13 : printf("Thirteen\n"); break;
                case 14 : printf("Fourteen\n"); break;
                case 15 : printf("Fifteen\n"); break;
                case 16 : printf("Sixteen\n"); break;
                case 17 : printf("Seventeen\n"); break;
                case 18 : printf("Eighteen\n"); break;
                case 19 : printf("Nineteen\n"); break;
                case 20 ... 29 :
                    printf("Twenty ");
                    n = num % 10;
                    goto onedigit;
                    break;
                case 30 ... 39 :
                    printf("Thirty ");
                    n = num % 10;
```

```
chamal@MacBook-Air: ~/Desktop $ ./a.out
Nine Hundred Ninety Nine
chamal@MacBook-Air: ~/Desktop $
```

```c
                n = num % 10;
                goto onedigit;
                break;
            case 40 ... 49 :
                printf("Fourty ");
                n = num % 10;
                goto onedigit;
                break;
            case 50 ... 59 :
                printf("Fifty ");
                n = num % 10;
                goto onedigit;
                break;
            case 60 ... 69 :
                printf("Sixty ");
                n = num % 10;
                goto onedigit;
                break;
            case 70 ... 79 :
                printf("Seventy ");
                n = num % 10;
                goto onedigit;
                break;
            case 80 ... 89 :
                printf("Eighty ");
                n = num % 10;
                goto onedigit;
                break;
            case 90 ... 99 :
                printf("Ninety ");
                n = num % 10;
                 goto onedigit;
                  break;
        }
    case 1:
        n = num;
onedigit:
        switch (n) {
            case 0 :
                switch (len) {
                    case 1: printf("Zero\n"); break;
                }
            case 1 : printf("One\n"); break;
            case 2 : printf("Two\n"); break;
            case 3 : printf("Three\n"); break;
            case 4 : printf("Four\n"); break;
            case 5 : printf("Five\n"); break;
            case 6 : printf("Six\n"); break;
            case 7 : printf("Seven\n"); break;
            case 8 : printf("Eight\n"); break;
            case 9 : printf("Nine\n"); break;
        }
    }

    return 0;
}
```

```
chamal@MacBook-Air: ~/Desktop $ ./a.out
Nine Hundred Ninety Nine
chamal@MacBook-Air: ~/Desktop $
```

(03)

(a)

```c
#include <stdio.h>

int main() {
    int n = 1;
    while (n <= 50) {
        printf("%i ", n);
        ++n;
    }
    printf("\n");
    return 0;
}
```

```
chamal@MacBook-Air: ~/Desktop $ ./a.out
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 3
3 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50
chamal@MacBook-Air: ~/Desktop $
```

(b)

```c
#include <stdio.h>

int main() {
    int n = 1;
    while (n <= 32) {
        printf("$ ");
        ++n;
    }
    printf("\n");
    return 0;
}
```

```
chamal@MacBook-Air: ~/Desktop $ ./a.out
$ $ $ $ $ $ $ $ $ $ $ $ $ $ $ $ $ $ $ $ $ $ $ $ $ $ $ $ $ $ $ $
chamal@MacBook-Air: ~/Desktop $
```

(c)

```c
#include <stdio.h>

int main() {
    int n = 10;
    while (n >= 0) {
        printf("%d ", n);
        --n;
    }
    printf("\n");
    return 0;
}
```

```
chamal@MacBook-Air: ~/Desktop $ ./a.out
10 9 8 7 6 5 4 3 2 1 0
chamal@MacBook-Air: ~/Desktop $
```

(d)

```c
#include <stdio.h>

int main() {
    double n = 10.0;
    while (n >= -5.0) {
        printf("%.2f ", n);
        n = n - 0.1;
    }
    printf("\n");
    return 0;
}
```

```
chamal@MacBook-Air: ~/Desktop $ ./a.out
10.00 9.90 9.80 9.70 9.60 9.50 9.40 9.30 9.20 9.10 9.00 8.90 8.80 8.70 8.60 8.50 8.40 8.
30 8.20 8.10 8.00 7.90 7.80 7.70 7.60 7.50 7.40 7.30 7.20 7.10 7.00 6.90 6.80 6.70 6.60
6.50 6.40 6.30 6.20 6.10 6.00 5.90 5.80 5.70 5.60 5.50 5.40 5.30 5.20 5.10 5.00 4.90 4.8
0 4.70 4.60 4.50 4.40 4.30 4.20 4.10 4.00 3.90 3.80 3.70 3.60 3.50 3.40 3.30 3.20 3.10 3
.00 2.90 2.80 2.70 2.60 2.50 2.40 2.30 2.20 2.10 2.00 1.90 1.80 1.70 1.60 1.50 1.40 1.30
 1.20 1.10 1.00 0.90 0.80 0.70 0.60 0.50 0.40 0.30 0.20 0.10 0.00 -0.10 -0.20 -0.30 -0.4
0 -0.50 -0.60 -0.70 -0.80 -0.90 -1.00 -1.10 -1.20 -1.30 -1.40 -1.50 -1.60 -1.70 -1.80 -1
.90 -2.00 -2.10 -2.20 -2.30 -2.40 -2.50 -2.60 -2.70 -2.80 -2.90 -3.00 -3.10 -3.20 -3.30
-3.40 -3.50 -3.60 -3.70 -3.80 -3.90 -4.00 -4.10 -4.20 -4.30 -4.40 -4.50 -4.60 -4.70 -4.8
0 -4.90 -5.00
chamal@MacBook-Air: ~/Desktop $
```

(e)

```c
#include <stdio.h>

int main() {
    int n = 10;
    int isFirst = 0;
    while (n <= 99) {
        printf("%d", n);
        isFirst = 0;
        if (n % 2 == 0) {
            if (!isFirst) {
                printf("\t-multiple of 2");
                isFirst = 1;
            } else {
                printf("\tmultiple of 2");
            }
        }
        if (n % 3 == 0) {
            if (!isFirst) {
                printf("\t-multiple of 3");
                isFirst = 1;
            } else {
                printf("\tmultiple of 3");
            }
        }
        if (n % 5 == 0) {
            if (!isFirst) {
                printf("\t-multiple of 5");
                isFirst = 1;
            } else {
                printf("\tmultiple of 5");
            }
        }
        printf("\n");
        ++n;
    }
    return 0;
}
```

```
61
62	-multiple of 2
63	-multiple of 3
64	-multiple of 2
65	-multiple of 5
66	-multiple of 2	multiple of 3
67
68	-multiple of 2
69	-multiple of 3
70	-multiple of 2	multiple of 5
71
72	-multiple of 2	multiple of 3
73
74	-multiple of 2
75	-multiple of 3	multiple of 5
76	-multiple of 2
77
78	-multiple of 2	multiple of 3
79
80	-multiple of 2	multiple of 5
81	-multiple of 3
82	-multiple of 2
83
84	-multiple of 2	multiple of 3
85	-multiple of 5
86	-multiple of 2
87	-multiple of 3
88	-multiple of 2
89
90	-multiple of 2	multiple of 3	multiple of 5
91
92	-multiple of 2
93	-multiple of 3
94	-multiple of 2
95	-multiple of 5
96	-multiple of 2	multiple of 3
97
98	-multiple of 2
99	-multiple of 3
```

(f)

```c
#include <stdio.h>

int main() {
    char n = 'a';
    while (n <= 122) {
        printf("%c ", n);
        ++n;
    }
    printf("\n");
    return 0;
}
```

```
chamal@MacBook-Air: ~/Desktop $ ./a.out
a b c d e f g h i j k l m n o p q r s t u v w x y z
chamal@MacBook-Air: ~/Desktop $
```

(04)

(a)

```c
#include <stdio.h>

int main() {
    for (int i = 20; i <= 35; i++) {
        printf("%d ", i);
    }

    printf("\n");
    return 0;
}
```

```
chamal@MacBook-Air: ~/Desktop $ ./a.out
20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35
chamal@MacBook-Air: ~/Desktop $
```

(b)

```c
#include <stdio.h>

int main() {
    for (int i = 1; i <= 11; i++) {
        printf("*");
    }

    printf("\n");
    return 0;
}
```

```
chamal@MacBook-Air: ~/Desktop $ ./a.out
***********
chamal@MacBook-Air: ~/Desktop $
```

(c)

i.

```c
#include <stdio.h>

int main() {
    for (int i = 1; i <= 5; i++) {
        printf("*\n");
    }

    return 0;
}
```

```
chamal@MacBook-Air: ~/Desktop $ ./a.out
*
*
*
*
*
chamal@MacBook-Air: ~/Desktop $
```

ii.

```c
#include <stdio.h>

int main() {
    for (int i = 1; i <= 5; i++) {
        for (int j = 1; j <=8; j++) {
            printf("*");
        }
        printf("\n");
    }

    return 0;
}
```

```
chamal@MacBook-Air: ~/Desktop $ ./a.out
********
********
********
********
********
chamal@MacBook-Air: ~/Desktop $
```

iii.

```c
#include <stdio.h>

int main() {
    for (int i = 1; i <= 5; i++) {
        for (int j = 1; j <= i; j++) {
            printf("*");
        }
        printf("\n");
    }

    return 0;
}
```

```
chamal@MacBook-Air: ~/Desktop $ ./a.out
*
**
***
****
*****
chamal@MacBook-Air: ~/Desktop $
```

iv.

```c
#include <stdio.h>

int main() {
    for (int i = 1; i <= 9; i++) {
        if (i <= 5) {
            for (int j = 1; j <= i; j++) {
                printf("*");
            }
        } else {
            for (int j = 1; j <= 10 % i; j++) {
                printf("*");
            }
        }
        printf("\n");
    }

    return 0;
}
```

```
chamal@MacBook-Air: ~/Desktop $ ./a.out
*
**
***
****
*****
****
***
**
*
chamal@MacBook-Air: ~/Desktop $
```

v.

```c
#include <stdio.h>

int main() {
    for (int i = 1; i <= 9; i++) {
        if (i <= 5) {
            for (int j = 1; j <= 5 - i; j++) {
                printf(" ");
            }
            for (int j = 1; j <= ((i * 2) - 1); j++) {
                printf("*");
            }
        } else {
            for (int j = 1; j <= 5 - (10 % i); j++) {
                printf(" ");
            }
            for (int j = 1; j <= (((10 % i) * 2) - 1); j++) {
                printf("*");
            }
        }
        printf("\n");
    }

    return 0;
}
```

```
chamal@MacBook-Air: ~/Desktop $ ./a.out
    *
   ***
  *****
 *******
*********
 *******
  *****
   ***
    *
chamal@MacBook-Air: ~/Desktop $
```

vi.

```c
#include <stdio.h>

int main() {
    int sum = 0;

    for (int i = 1; i <= 9; i++) {
        if (i <= 5) {
            sum = sum + i;
            for (int j = 1; j <= i; j++) {
                printf("*");
            }
            printf("\t%d", sum);
        } else {
            for (int j = 1; j <= 10 % i; j++) {
                printf("*");
            }
        }
        printf("\n");
    }

    return 0;
}
```

```
chamal@MacBook-Air: ~/Desktop $ ./a.out
*       1
**      3
***     6
****    10
*****   15
****
***
**
*
chamal@MacBook-Air: ~/Desktop $
```

vii.

```c
#include <stdio.h>

int main() {
    for (int i = 1; i <= 5; i++) {
        for (int j = 1; j <= i - 1; j++) {
            printf(" ");
        }
        for (int j = 1; j <= 8; j++) {
            printf("*");
        }
        printf("\n");
    }

    return 0;
}
```

```
chamal@MacBook-Air: ~/Desktop $ ./a.out
********
 ********
  ********
   ********
    ********
chamal@MacBook-Air: ~/Desktop $
```

(05)

```c
#include <stdio.h>

int main() {
    double d;
    float f;
    long l;
    int i;

    i = l = f = d = 100/3;
    printf("%d\t%ld\t%f\t%f\n",i,l,f,d);

    d = f = l = i = 100/3;
    printf("%d\t%ld\t%f\t%f\n",i,l,f,d);

    i = l = f = d = 100/3;
    printf("%d\t%ld\t%f\t%f\n",i,l,f,d);

    d = f = l = i = (float)100/3;
    printf("%d\t%ld\t%f\t%f\n",i,l,f,d);

    i = l = f = d = (double)(100000/3);
    printf("%d\t%ld\t%f\t%f\n",i,l,f,d);

    d = f = l = i = 100000/3;
    printf("%d\t%ld\t%f\t%f\n",i,l,f,d);

    return 0;
}
```

```
chamal@MacBook-Air: ~/Desktop $ ./a.out
33      33      33.000000       33.000000
33      33      33.000000       33.000000
33      33      33.000000       33.000000
33      33      33.000000       33.000000
33333   33333   33333.000000    33333.000000
33333   33333   33333.000000    33333.000000
chamal@MacBook-Air: ~/Desktop $
```

(06)

```c
#include <stdio.h>

int main () {
    double d = 3.2, x;
    int i = 2, y;

    x = (y = d/i) *2;
    printf("%f\t%d\n", x, y);

    y = (x = d/i)*2;
    printf("%f\t%d\n", x, y);

    y = d * (x=2.5/d);
    printf("%f\n",y);

    x=d * (y= ((int)2.9+1.1)/d);
    printf("%f\t%d\n", x, y);
}
```

```
chamal@MacBook-Air: ~/Desktop $ ./a.out
2.000000        1
1.600000        3
0.000000
0.000000        0
chamal@MacBook-Air: ~/Desktop $
```